Theory of Computation 1-7 Review of Mathematical Theory Let us discuss each one by one

Negation

horizontal bar on letter. S is a proposition then S defines NOT s. The truth table for this connective is

S	1s
T	F.
F	T

) Conjunction (AND)

If A and B are two propositions then conjunction of A and B is denoted by A^B.The truth table for this connective is

A	B	A^B
T	T	T
T	F	F
F	T	F
F	F	F

Disjunction OR

If A and B are two propositions then disjunction of A and B is denoted by AvB.The truth table for this connective is

A	B	AvB
T	T	T
T	F	T

A	B	AvB
F	T	
		T
F	F	F

Iv) Implication

A and B are two propositions then if A then B is a proposition.It can be denoted by A-B or A=B.

6 8 9
0
R I
G H

M

comma

Theory of Computation 1-8 Revlew of Mathematical Theory The truth table for this connective is

A	B	A-B
T	T	T
T	F	F
F	T	T
F	F	T

For example: Find truth value for following proposition. If an apple is not green then it must be red". Let A denotes An apple is not green and B denotes It must be red". The statement A is false and B is true. Hence the proposition is true. vIf and only if If A and B are two propositions then A if and only if B. It can be denoted as A The truth table will be

	B	AB
T	T	T
T	F	F
F	T	F
F	F	T

For example I will go to the chool if and only if it is not raining. Here statement A proposition is false.

1.5 Relations

forma pair based on this certain relationship.

elements.

pair is chosen from a set called range

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Properties of Relations

A relation R on set S is

1. Reflexive if iRi for all i in S.
2. Irreflexive if iRi is false for all i in S.
3. Transitive if iRj and jRk imply iRk
4. Symmetric if iRj implies jRi.
5. Asymmetric if iRj implies that jRi is false.

Every asymmetric relation must be irreflexive.

For example: if A=ab| then

reflexive relation R can be=|a.a.b, b1

irreflexive relation R can be =|(a, b

transitive relation R can be=ab)b, aaa

symmetric relation R can be =|a.b)b, a

asymmetric relation R can be =(a b)

Equivalent Relation

A relation is said to be equivalence relation if it is reflexive, symmetric and transitive, over some set S. Suppose R is a set of relations and S is the set of elements.

For example: S is the set of lines in a plane and R is the relation of lines intersecting to each other. A relation is said to be antisymmetric if a, b) e R and b a)e R then a =a.

Example 1.5: Determine whether R is equivalence relation or not where A=10121R=0010112, 2)2, 1 SolutionThe R is reflexive because 0, 0).1, 1).2.2e R. Where as R is not symmetric because (0. 1) is not in R whereas (1, 0) is in R. Hence the R is not a equivalence relation.

Closures of Relations

Sometimes when the relation R is given, it may not be reflexive or transitive By adding some pairs we make the relation as reflexive or transitive.

For exampleLetab)bc)aabb